



HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

FUNDAMENTALS OF OPTIMIZATION

Integer Linear Programming

ONE LOVE. ONE FUTURE.

- Introduction to integer programming
 - Relaxation and bound
 - Branch and bound
 - Cutting plane
 - Integer rounding
 - Gomory cut
 - Schema of branch-and-cut algorithm
- Some techniques to linearize a constraint
- Exercises
 - Blood type correction problem
 - TSP

RELAXATION AND BOUND

- Given an Integer Program (IP)
- Find decreasing sequence of upper bounds

$$\bar{z}_1 > \bar{z}_1 > \dots > \bar{z}_s \geq z$$

- Find increasing sequence of lower bounds

$$\underline{z}_1 < \underline{z}_1 < \dots < \underline{z}_t \leq z$$

- Algorithm stop when $\bar{z}_s - \underline{z}_t \leq \varepsilon$

RELAXATION AND BOUND

- Primal bounds
 - Every feasible solution $x^* \in X$ provides a lower bound of the maximization problem: $\underline{z} = cx^* \leq z$
 - Example: in TSP, every closed tour is a upper bound of the objective function (as TSP is a minimization problem)

RELAXATION AND BOUND

- Dual bounds
 - Finding upper bounds for a maximization problem (or lower bounds for a minimization problem) gives dual bounds of the objective
- **Definition** A problem $(RP) z^R = \max\{f(x): x \in T \subseteq R^n\}$ is a relaxation of $(IP) z = \max\{c^T x: x \in X \subseteq Z^n\}$ if:
 - $X \subseteq T$
 - $f(x) \geq c^T x, \forall x \in X$

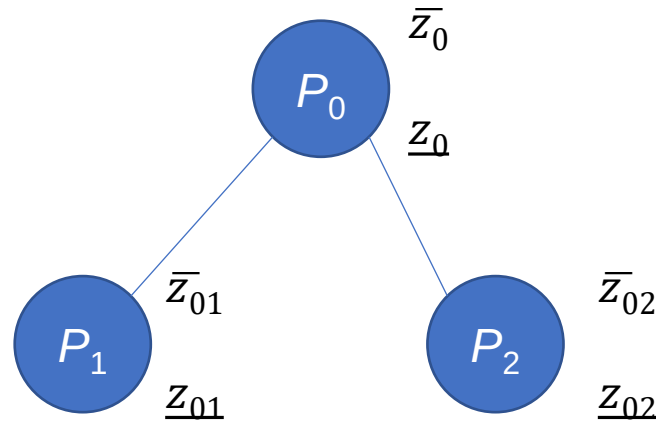
RELAXATION AND BOUND

- Linear Relaxation

- $Z^{LP} = \max\{c^T x : x \in P\}$ with $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is a linear relaxation program of the (IP) $\max\{c^T x : x \in P \cap \mathbb{Z}^n\}$

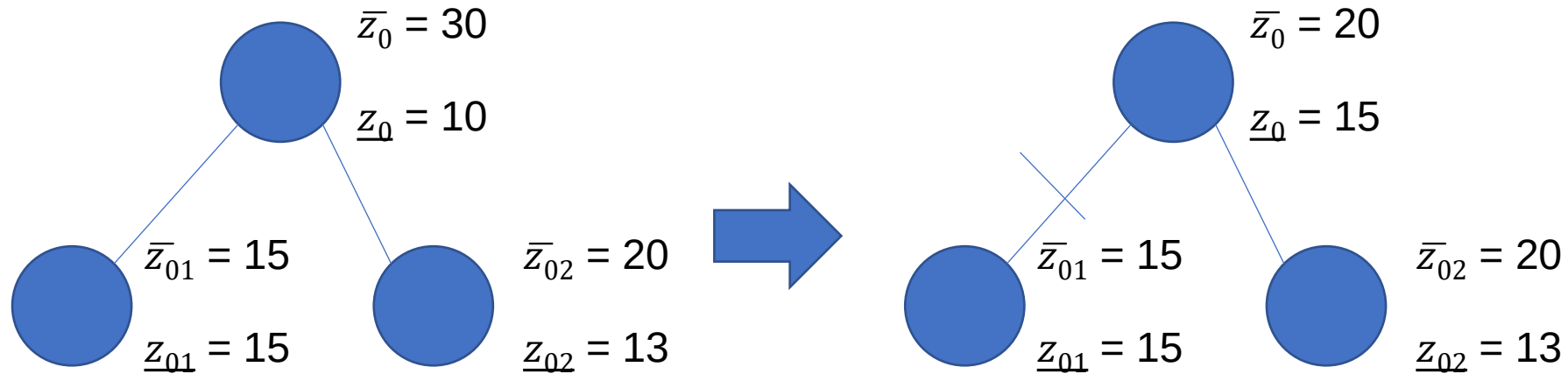
BRANCH AND BOUND

- Feasible region of P_0 is divided into feasible regions of P_1 and P_2 : $X(P_0) = X(P_1) \cup X(P_2)$



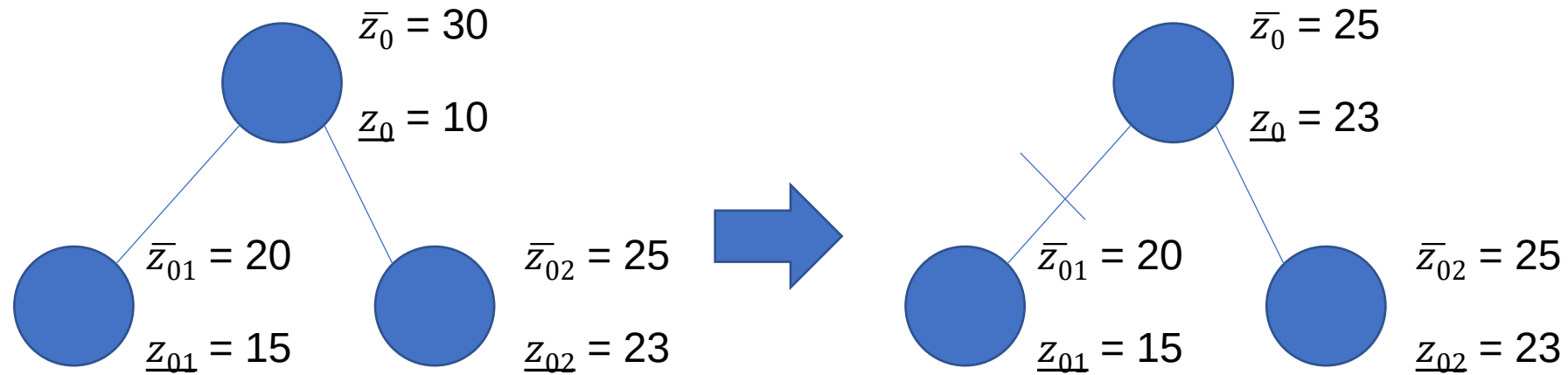
BRANCH AND BOUND

- Feasible region of P_0 is divided into feasible regions of P_1 and P_2 : $X(P_0) = X(P_1) \cup X(P_2)$



BRANCH AND BOUND

- Feasible region of P_0 is divided into feasible regions of P_1 and P_2 : $X(P_0) = X(P_1) \cup X(P_2)$



BRANCH AND BOUND

- Branch and Bound schema

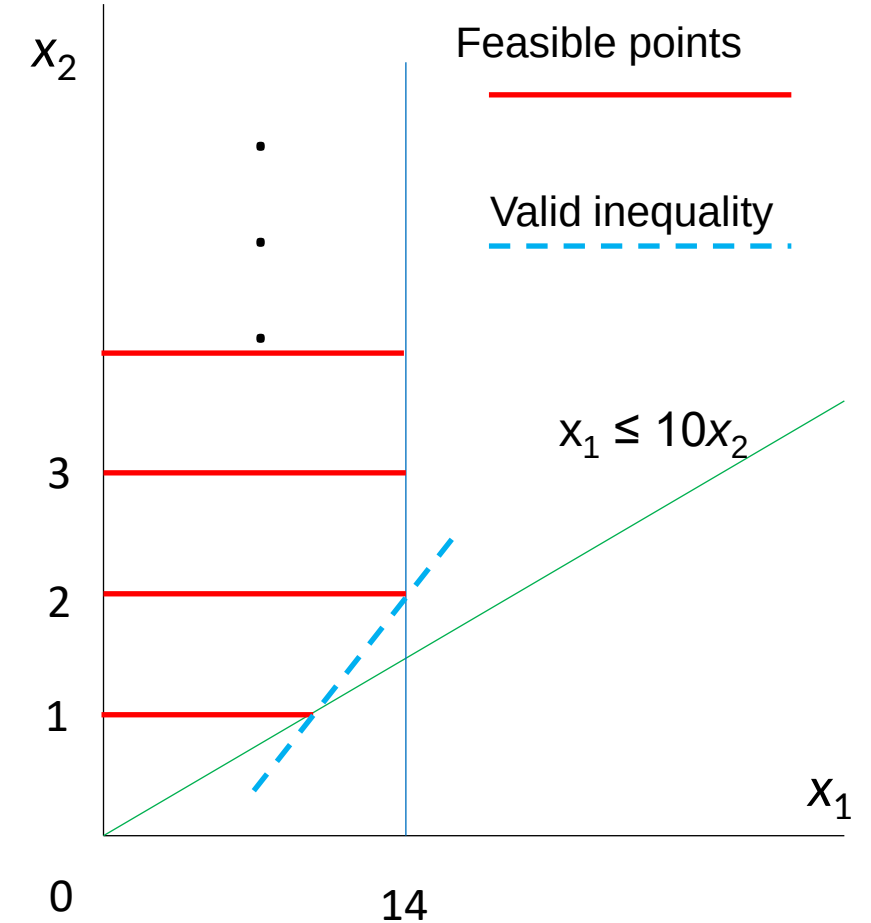
```
Initial problem  $S$  with formulation  $P$  on a list  $L$ ;  
Incumbent  $x^*$  is initialized with primal bound  $\underline{z} = -INF$ ;  
while  $L$  not empty do {  
    Select a problem  $S^i$  with formulation  $P^i$  from  $L$ ;  
    Solve LP relaxation over  $P^i$  got dual bound  $\bar{z}^i$  and solution  $x^i(LP)$ ;  
    if  $\bar{z}^i \leq \underline{z}$  then continue; // prune by dual bound;  
    if  $x^i(LP)$  integer then{  
         $\underline{z} = \bar{z}^i$  ;  
         $x^* = x^i(LP)$ ;  
    }else{  
        select a component  $x_i$  of  $x^i(LP)$  whose value  $\lambda_i$  is fractional;  
         $P_1^i = P^i \cup (x_i \leq \lfloor \lambda_i \rfloor)$ ,  $P_2^i = P^i \cup (x_i \geq \lceil \lambda_i \rceil)$ ;  
        add  $P_1^i$  and  $P_2^i$  to  $L$ ;  
    }  
}  
return  $x^*$ ;
```

CUTTING PLANE

- Given a (MIP) $\max\{c^T z: z \in X\}$
- Inequality $\pi z \leq \pi_0$ is called a valid inequality if $\pi z \leq \pi_0$ is true for all $z \in X$
- Finding valid inequalities allows us to narrow the search space, transform the (MIP) to a corresponding (LP) in which an optimal solution to (LP) is an optimal solution to the original (MIP)

CUTTING PLANE

- Example, consider a MIP with $X = \{(x_1, x_2): x_1 \leq 10x_2, 0 \leq x_1 \leq 14, x_2 \in \mathbb{Z}_+^1\}$
- Red lines represent X
- $x_1 \leq 6 + 4x_2$ is a valid inequality (dashed line)



CUTTING PLANE

- Example: **Integer Rounding**

- Consider feasible region $X = P \cap \mathbb{Z}^3$ where $P = \{x \in \mathbb{R}_+^3 : 5x_1 + 9x_2 + 13x_3 \geq 19\}$

- From $5x_1 + 9x_2 + 13x_3 \geq 19$ we have $x_1 + \frac{9}{5}x_2 + \frac{13}{5}x_3 \geq \frac{19}{5}$

$$\rightarrow x_1 + 2x_2 + 3x_3 \geq \frac{19}{5}$$

- As x_1, x_2, x_3 are integers, so we have

$$x_1 + 2x_2 + 3x_3 \geq \left\lceil \frac{19}{5} \right\rceil = 4 \text{ (this is a valid inequality for } X)$$

GOMORY CUT

- (IP) $\max \{cx: Ax = b, x \geq 0 \text{ and integer}\}$
- Solve corresponding linear programming relaxation (LP) $\max \{cx: Ax = b, x \geq 0\}$
- Suppose with an optimal basis, the (LP) is rewritten in the form

$$\overline{a_{00}} + \sum_{j \in JN} \overline{a_{0j}} x_j \rightarrow \max$$

$$x_{B_u} + \sum_{j \in JN} \overline{a_{uj}} x_j = \overline{a_{u0}}, u = 1, 2, \dots, m$$

$$x \geq 0 \text{ and integer}$$

with $\overline{a_{0j}} \leq 0$ (as these coefficients corresponds to a maximizer), and $\overline{a_{u0}} \geq 0$

GOMORY CUT

- (IP) $\max \{cx: Ax = b, x \geq 0 \text{ and integer}\}$
- Solve corresponding linear programming relaxation (LP) $\max \{cx: Ax = b, x \geq 0\}$
- Suppose with an optimal basis, the (LP) is rewritten in the form

$$\overline{a}_{00} + \sum_{j \in JN} \overline{a}_{0j} x_j \rightarrow \max$$

$$x_{B_u} + \sum_{j \in JN} \overline{a}_{uj} x_j = \overline{a}_{u0}, u = 1, 2, \dots, m$$

$$x \geq 0 \text{ and integer}$$

with $\overline{a}_{0j} \leq 0$ (as these coefficients corresponds to a maximizer), and $\overline{a}_{u0} \geq 0$

- If the basic optimal solution x^* is not integer, then there exists some row u with $\overline{a}_{u0}$ is not integer
 $\rightarrow$ Create a Gomory cut $x_{B_u} + \sum_{j \in JN} \lfloor \overline{a}_{uj} \rfloor x_j \leq \lfloor \overline{a}_{u0} \rfloor$ (1)

- **Example.** Consider the Integer Linear Program (ILP)

$$f(x_1, x_2, x_3, x_4, x_5) = x_1 + x_2 \rightarrow \max$$

$$2x_1 + x_2 + x_3 = 8$$

$$3x_1 + 4x_2 + x_4 = 24$$

$$x_1 - x_2 + x_5 = 2$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0 \text{ and integer}$$

- Solve the corresponding (LP)

$$f(x_1, x_2, x_3, x_4, x_5) = x_1 + x_2 \rightarrow \max$$

$$2x_1 + x_2 + x_3 = 8$$

$$3x_1 + 4x_2 + x_4 = 24$$

$$x_1 - x_2 + x_5 = 2$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

→ Optimal solution (1.6, 4.8, 0, 0, 5.2) with $J_B = (1, 2, 5)$ and $J_N = (3, 4)$

Rewrite the original (ILP)

$$f(x_1, x_2, x_3, x_4, x_5) = 6.4 - 0.2x_3 - 0.2x_4 \rightarrow \max$$

$$x_1 + 0.8x_3 - 0.2x_4 = 1.6$$

$$x_2 - 0.6x_3 + 0.4x_4 = 4.8$$

$$x_5 - 1.4x_3 + 0.6x_4 = 5.2$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0 \text{ and integer}$$

GOMORY CUT

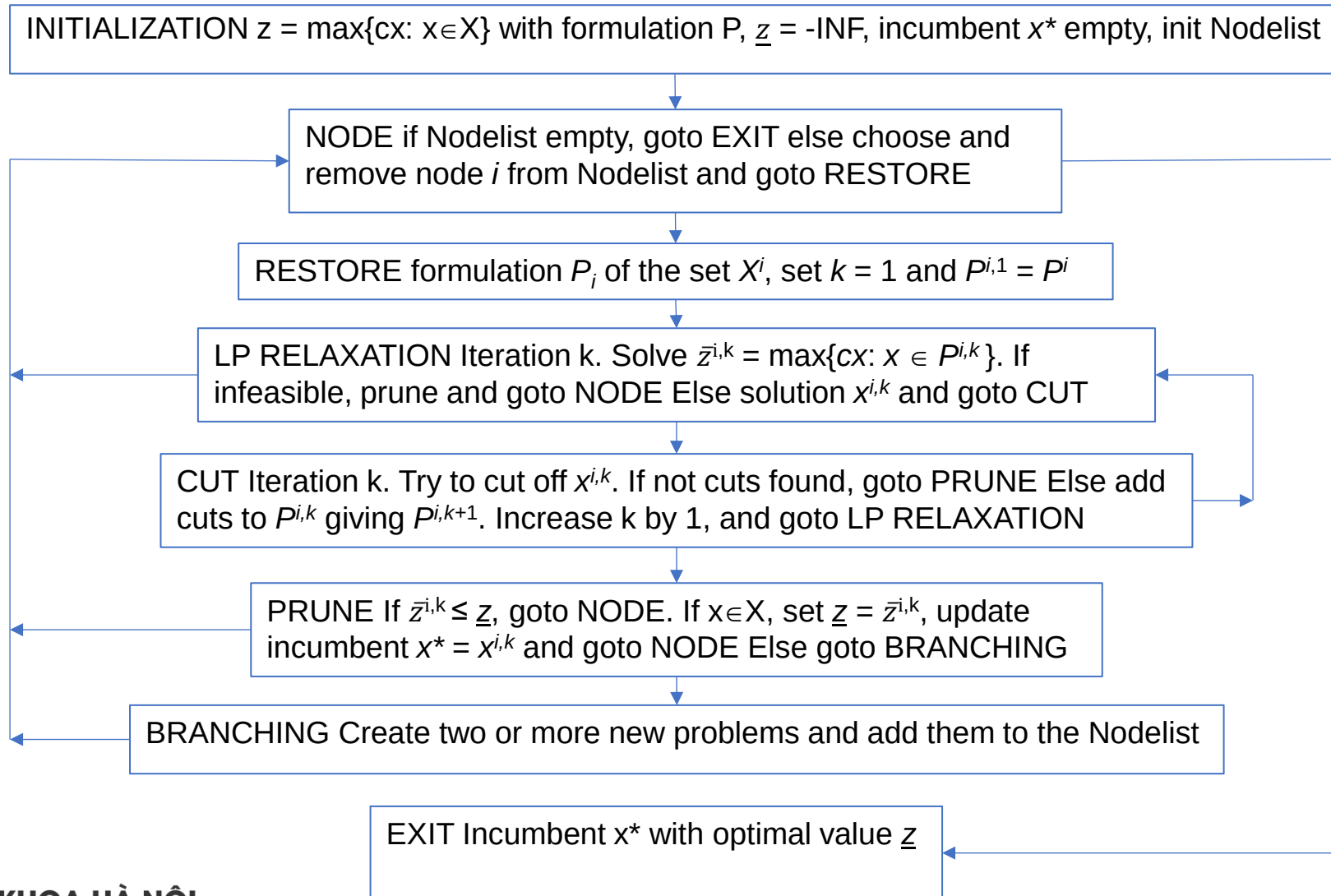
- Consider $x_1 + 0.8x_3 - 0.2x_4 = 1.6$
→ $x_1 + 0x_3 - x_4 \leq \lfloor 1.6 \rfloor = 1$
→ Add slack variable x_6 and constraint (Gomory cut): $0.8x_3 + 0.8x_4 - x_6 = 0.6$
- Consider $x_2 - 0.6x_3 + 0.4x_4 = 4.8$
→ $x_2 - x_3 + 0.4x_4 \leq \lfloor 4.8 \rfloor = 4$
→ Add slack variable x_7 and constraint (Gomory cut): $0.4x_3 + 0.4x_4 - x_7 = 0.8$
- Consider $x_5 - 1.4x_3 + 0.6x_4 = 5.2$
→ $x_5 - 2x_3 + 0x_4 \leq \lfloor 5.2 \rfloor = 5$
→ Add slack variable x_8 and constraint (Gomory cut) $0.6x_3 + 0.6x_4 - x_8 = 0.2$

GOMORY CUT

- Add Gomory cuts

- $x_1 + 0.8x_3 - 0.2x_4 = 1.6 \rightarrow \text{add Gomory cut: } x_1 + 0x_3 - x_4 \leq 1$
- $x_2 - 0.6x_3 + 0.4x_4 = 4.8 \rightarrow \text{add Gomory cut: } x_2 - x_3 + 0x_4 \leq 4$
- $x_5 - 1.4x_3 + 0.6x_4 = 5.2 \rightarrow \text{add Gomory cut: } x_5 - 2x_3 + 0x_4 \leq 5$

BRANCH AND CUT [Wolsey, 98]



- Introduction to integer programming
 - Relaxation and bound
 - Branch and bound
 - Cutting plane
 - Integer rounding
 - Gomory cut
 - Schema of branch-and-cut algorithm
- **Some techniques to linearize a constraint**
- Exercises
 - Blood type correction problem
 - TSP

SOME TECHNIQUES TO LINEARIZE A CONSTRAINT

- If then constraint
- Multiplication of Binary Variables
- Multiplication of Binary and Continuous Variables
- Maximum/Minimum Operators
- Absolute Value Function

If Then

- $x, y \in \{0, 1\}$ If $x = 1$ then $y = 1$ otherwise nothing
- Scenarios:
 - $x = 1, y = 1$
 - $x = 0, y = 0$
 - $x = 0, y = 1$
- Transformation:
 - $(x - 1) \times M + 1 \leq y$, in which $M \geq 2$
- Applications in TSP: x_{ij} is a binary variable to indicated that the salesman goes directly from the city i to the city j or not; and y_i is the city that the salesman visits at i^{th} time. The subtour : *if $x_{ij} = 1$ then $y_j = y_i + 1$*
 - $(1 - x_{ij}) \times M + y_j \geq y_i + 1$
 - $(x_{ij} - 1) \times M + y_j \leq y_i + 1$in which M is a very large number

Multiplication of Binary Variables

- Non-linear constraint: $z = x \times y$ with $x, y \in \{0, 1\}$

- Transformation:

- $z \leq x$
- $z \leq y$
- $x + y \leq z + 1$

x	y	$x \cdot y$	Constraints	ImPLY
0	0	0	$z \leq 0$ $z \leq 0$ $z \geq -1$ $z \in \{0, 1\}$	$z = 0$
0	1	0	$z \leq 0$ $z \leq 1$ $z \geq 0$ $z \in \{0, 1\}$	$z = 0$
1	0	0	$z \leq 1$ $z \leq 0$ $z \geq 0$ $z \in \{0, 1\}$	$z = 0$
1	1	1	$z \leq 1$ $z \leq 1$ $z \geq 1$ $z \in \{0, 1\}$	$z = 1$

MULTIPLICATION OF BINARY AND CONTINUOUS VARIABLES

- Non-linear constraint: $z = x \times y$ with $x \in \{0, 1\}, y \in R$
- Transformation:

- $z \leq y$
- $z \leq M \times x$
- $M(x - 1) + y \leq z$

x	y	$x \cdot y$	Constraints	Imply
0	$m : 0 \leq m \leq u$	0	$z \leq m$ $z \leq 0$ $z \geq m - u$ $z \geq 0$	$z = 0$
1	$m : 0 \leq m \leq u$	m	$z \leq m$ $z \leq u$ $z \geq m$ $z \geq 0$	$z = m$

MAXIMUM/MINIMUM OPERATORS

- Non-linear constraint: $z = \max(x, y)$ with $x, y \in R$
- Transformation:
 - $z \geq y$
 - $z \geq x$
 - $z \leq M \times t_1 + x$
 - $z \leq M \times t_2 + y$
 - $t_1 + t_2 \leq 1$

ABSOLUTE VALUE FUNCTION

- Absolute constraint: $|f(x)| \leq z$
- Transformation:
 - $f(x) \leq z$
 - $-f(x) \leq z$
- Absolute constraint: $|f(x)| \geq z$
- Transformation
 - $f(x) \geq z - t \times M$
 - $f(x) \leq -z + (1 - t) \times M$
- Absolute constraint: $|f(x)| = z$
- Transformation
 - $f(x) = z - 2 \times t \times z$

- Introduction to integer programming
 - Relaxation and bound
 - Branch and bound
 - Cutting plane
 - Integer rounding
 - Gomory cut
 - Schema of branch-and-cut algorithm
- Some techniques to linearize a constraint
- **Exercises**
 - Blood type correction problem
 - TSP

BLOOD TYPE CORRECTION PROBLEM

- In Vietnam's National Resident Database, information about each citizen's blood type is recorded. Due to the manual data collection process, errors exist in some records within the database. The objective of this exercise is to determine the minimum number of records that must be corrected to ensure the database's accuracy and consistency. The problem is defined as follows: Consider a family with multiple individuals, where each individual is assigned a blood type (O, A, B, or AB) and linked to two parents. Note that information about an individual's parents may sometimes be missing. Using the rules of blood type inheritance and the provided family information, answer the question: What is the smallest number of individuals in the family whose blood types must be changed to ensure that the blood type inheritance rules are satisfied for the entire family?
- Question 1: Propose Mixed Integer Program for the problem
- Question 2: Implement the MIP in Hustack (programming.daotao.ai)

Blood types of parent	Possible blood types of the child
A, A	O, A
B, B	O, B
O, O	O
A, B	O, A, B, AB
A, O	A, O
A, AB	A, B, AB
B, AB	A, B, AB
O, B	O, B
O, AB	A, B
AB, AB	A, B, AB

INTEGER PROGRAM FOR BLOOD TYPE CORRECTION PROBLEM

- From the blood type combination rules presented, we derive the following four rules. Given three individuals X , Y , and Z , where X and Y are the parents of Z :
- **Rule 1:** If the blood type of Z is O, then the blood types of both X and Y must not be AB.
- **Rule 2:** If the blood type of Z is A, then the blood type of either X or Y must be A or AB.
- **Rule 3:** If the blood type of Z is B, then the blood type of either X or Y must be B or AB.
- **Rule 4:** If the blood type of Z is AB, then the blood type of either X or Y must be AB and both X and Y are different from O; or the blood types of X and Y must be A and B; or B and A, respectively.

Blood types of parent	Possible blood types of the child
A, A	O, A
B, B	O, B
O, O	O
A, B	O, A, B, AB
A, O	A, O
A, AB	A, B, AB
B, AB	A, B, AB
O, B	O, B
O, AB	A, B
AB, AB	A, B, AB

DECISION VARIABLES

- For each individual i ($i \in \{1, \dots, n\}$), we define five binary variables $x_i^1, x_i^2, x_i^3, x_i^4$, and y_i , where:
 - * x_i^1 (resp. x_i^2, x_i^3 , and x_i^4) is equal to 1 if their blood type is O (resp. A, B, and AB), and is equal to 0 otherwise;
 - * y_i is equal to 1 if the blood type of individual i must be changed.
- For each married couple $\langle i, j \rangle$, we define a pair of binary variables z_{ij} and z_{ji} , where:
 - * $z_{ij} = 1$ if the blood type of i is A and the blood type of j is B; otherwise, $z_{ij} = 0$.
 - * $z_{ji} = 1$ if the blood type of i is B and the blood type of j is A; otherwise, $z_{ji} = 0$.

CONSTRAINTS

$$x_i^1 = 1, \forall i \in \{1, \dots, n\}, l_i = 1, b_i = O \quad (1)$$

$$x_i^2 = 1, \forall i \in \{1, \dots, n\}, l_i = 1, b_i = A \quad (2)$$

$$x_i^3 = 1, \forall i \in \{1, \dots, n\}, l_i = 1, b_i = B \quad (3)$$

$$x_i^4 = 1, \forall i \in \{1, \dots, n\}, l_i = 1, b_i = AB \quad (4)$$

$$x_i^1 + x_i^2 + x_i^3 + x_i^4 = 1, \forall i \in \{1, \dots, n\} \quad (5)$$

$$z_{ij} \leq x_i^1, \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$z_{ij} \leq x_j^2, \forall i, j \in \{1, \dots, n\}, \quad (7)$$

$$x_i^1 + x_j^2 \leq z_{ij} + 1, \forall i, j \in \{1, \dots, n\}, \quad (8)$$

$$z_{ji} \leq x_i^2, \forall i, j \in \{1, \dots, n\}, \quad (9)$$

$$z_{ji} \leq x_j^1, \forall i, j \in \{1, \dots, n\}, \quad (10)$$

$$x_i^2 + x_j^1 \leq z_{ji} + 1, \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$2 \times (1 - x_i^1) - x_{m_i}^4 - x_{f_i}^4 \geq 0, \forall i, f_i, m_i \in \{1, \dots, n\} \quad (12)$$

$$(1 - x_i^2) + x_{m_i}^2 + x_{f_i}^2 + x_{m_i}^4 + x_{f_i}^4 \geq 1, \forall i, f_i, m_i \in \{1, \dots, n\} \quad (13)$$

$$(1 - x_i^3) + x_{m_i}^3 + x_{f_i}^3 + x_{m_i}^4 + x_{f_i}^4 \geq 1, \forall i, f_i, m_i \in \{1, \dots, n\} \quad (14)$$

$$(1 - x_i^4) + x_{m_i}^4 + x_{f_i}^4 - (x_{m_i}^1 + x_{f_i}^1) + z_{m_i f_i} + z_{f_i m_i} \geq 1, \forall i, f_i, m_i \in \{1, \dots, n\} \quad (15)$$

$$y_i = x_i^2 + x_i^3 + x_i^4, \forall i \in \{1, \dots, n\}, b_i = O \quad (16)$$

$$y_i = x_i^1 + x_i^3 + x_i^4, \forall i \in \{1, \dots, n\}, b_i = A \quad (17)$$

$$y_i = x_i^1 + x_i^2 + x_i^4, \forall i \in \{1, \dots, n\}, b_i = B \quad (18)$$

$$y_i = x_i^1 + x_i^2 + x_i^3, \forall i \in \{1, \dots, n\}, b_i = AB \quad (19)$$

CONSTRAINTS

- Constraints (1), (2), (3), and (4) indicate that if the correctness level of individual i is 1, then their blood type is fixed.
- Constraints (5) ensure that each individual has only one blood type.
- Constraints (6), (7), and (8) ensure that $z_{ij} = 1$ if and only if $x_i^1 = 1$ and $x_j^2 = 1$. This means that if the blood type of individual i is A and the blood type of individual j is B, then $z_{ij} = 1$.
- Constraints (9), (10), and (11) ensure that $z_{ji} = 1$ if and only if $x_i^2 = 1$ and $x_j^1 = 1$. This means that if the blood type of individual i is B and the blood type of individual j is A, then $z_{ji} = 1$.
- Constraints (12) ensure that if the blood type of individual i is O ($x_i^1 = 1$), then the blood types of both their parents must not be AB ($x_{m_i}^4 = x_{f_i}^4 = 0$). These represent **Rule 1**.
- Constraints (13) ensure that if the blood type of individual i is A ($x_i^2 = 1$), then the blood type of either their father or mother is either A or AB ($x_{m_i}^2 + x_{f_i}^2 + x_{m_i}^4 + x_{f_i}^4 \geq 1$). These represent **Rule 2**.

CONSTRAINTS

- Constraints (14) ensure that if the blood type of individual i is B ($x_i^3 = 1$), then the blood type of at least one of the two parents is either B or AB ($x_{m_i}^3 + x_{f_i}^3 + x_{m_i}^4 + x_{f_i}^4 \geq 1$). These represent **Rule 3**.
- Constraints (15) and (5) ensure that if the blood type of individual i is AB ($x_i^4 = 1$), then the blood type of at least one of the two parents is AB and the blood type of no one is O ; or blood types of two parents are A and B . These represent **Rule 4**.
- Constraints (16), (17), (18), and (19) state that if the blood type of individual i is modified, then the value of the penalty variable y_i must be equal to 1.
- The objective function of the model is to minimize the number of changes.

$$\text{Minimize } \sum_{i=1}^n y_i$$

- Introduction to integer programming
 - Relaxation and bound
 - Branch and bound
 - Cutting plane
 - Integer rounding
 - Gomory cut
 - Schema of branch-and-cut algorithm
- Some techniques to linearize a constraint
- **Exercises**
 - Blood type correction problem
 - TSP

Traveling Salesman Problem

- Input:
 - n number of cities
 - $c(i, j)$ traveling cost from the city i to the city j
- Variables:
 - Binary variable x_{ij} with $i, j \in \{1, \dots, n\}, i \neq j$ is equal to 1 if the traveler go from the city i to the city j in the optimal plan, otherwise the value of this variable is equal to 0;
- Constraints:
 - For each city, the traveler goes in once and goes out once
$$\sum_{j \in \{1, \dots, n\}} x_{ij} = \sum_{j \in \{1, \dots, n\}} x_{ji} = 1, \forall i \in \{1, \dots, n\}$$
 - No subtour
$$\sum_{i, j \in S, i \neq j} x_{ij} \leq |S| - 1, \forall S \subset \{1, \dots, n\}$$
- Objective: *Minimize* $\sum_{i, j \in \{1, \dots, n\}, i \neq j} c(i, j) x_{ij}$

Remove them?



A large graphic on the left side of the slide. It features a dark blue background with a circular pattern of red dots of varying sizes, creating a sense of depth and movement. The word "HUST" is centered within this graphic in a white, bold, sans-serif font.

HUST

THANK YOU !